

SSC CGL Quant Formula Recall Sheet

SSC CGL / Tier Prep / Quantitative Aptitude

Premium Snapshot

Exam: SSC CGL

Subject: Quantitative Aptitude

Topic: Arithmetic Geometry Mensuration Trigonometry

This Study Hub premium pack is a guided sheet. Keep official syllabus, official notices, and standard books as the source of truth.

Study Flow

- Step 1: Open the matching syllabus or textbook folder first.
- Step 2: Convert this sheet into a one-page note and a short practice set.
- Step 3: Solve PYQ, mock, or board-style questions from the same folder family.
- Step 4: Add every mistake to one error log and revise it weekly.

Premium Content

- Keep this sheet as the final 20-minute formula recall loop before timed quant practice.
- Do not only read formulas. Convert every formula into one mental example so recall becomes active.
- Percentage: $x\%$ of $y = y\%$ of x . Use it when one side is easier, such as 18% of $50 = 50\%$ of $18 = 9$.
- Successive percentage change: $a\%$ and $b\%$ gives $a+b+ab/100$. For decrease, use negative sign.
- Profit percent = $\text{profit}/\text{CP} \times 100$. Loss percent = $\text{loss}/\text{CP} \times 100$. Discount percent = $\text{discount}/\text{MP} \times 100$.
- SI = $\text{PRT}/100$. CI for 2 years = $2r + r^2/100$ percent. CI for 3 years = $3r + 3r^2/100 + r^3/10000$ percent.
- Ratio split: if total is T and ratio is $a:b$, parts are $T \cdot a/(a+b)$ and $T \cdot b/(a+b)$.
- Average: $\text{new average} = \text{old average} + \text{total change} / \text{number of items}$.
- Time-work: efficiency is work per day. If A does work in x days, A efficiency = $1/x$.
- Pipes and cistern: inlet positive, outlet negative. Net work = sum of individual efficiencies.
- Speed: $\text{distance} = \text{speed} \times \text{time}$. Train crossing pole = $\text{train length} / \text{speed}$. Train crossing platform = $(\text{train} + \text{platform length}) / \text{speed}$.
- Boats: downstream speed = $\text{boat} + \text{stream}$. Upstream speed = $\text{boat} - \text{stream}$.
- Simple interest vs compound interest for 2 years: $\text{CI} - \text{SI} = P \cdot r^2/10000$.
- Algebra: $(a+b)^2$, $(a-b)^2$, a^2-b^2 , $(a+b)^3$, $(a-b)^3$, a^3+b^3 , a^3-b^3 should be instant.
- Geometry: triangle area = $1/2 bh$. Equilateral triangle area = $\sqrt{3}/4 a^2$. Inradius area relation: $\text{area} = r \cdot s$.
- Circle: circumference = $2\pi r$, area = πr^2 , sector area = $\theta/360 \cdot \pi r^2$, arc length = $\theta/360 \cdot 2\pi r$.
- Mensuration: cylinder volume = $\pi r^2 h$, cone volume = $1/3 \pi r^2 h$, sphere volume = $4/3 \pi r^3$.
- Trigonometry values: $\sin/\cos/\tan$ for $0, 30, 45, 60, 90$ degrees must be recalled without writing table.
- Trig identities: $\sin^2 A + \cos^2 A = 1$, $\sec^2 A - \tan^2 A = 1$, $\text{cosec}^2 A - \cot^2 A = 1$.
- Practice rule: after every mock, add the missing formula to this recall list and revise it next morning.

Output Checklist

- One syllabus map or chapter map is complete.
- One practice set is attempted in timed mode.
- Wrong answers are grouped by concept, careless error, or time pressure.
- Next revision date is written before closing this pack.